

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1787

Forman 100 kV
May 25, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1787. The Interconnection Customer proposes a 107.0 MW battery energy storage facility interconnecting at the Forman 100 kV substation in the Illinois area. The request was received on June 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 439 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,209,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Arch Smith – Sweatt 100 KV
Contingency: Sweatt – Arch Smith 100 KV Ckt 2
Loading: 162.9% (Rating: 241 MVA)
DFAX: 7.8%
Pre-project loading: 159.4%
2. Facility: Sweatt – Arch Smith 100 KV #2
Contingency: Sweatt – Arch Smith 100 KV L
Loading: 162.9% (Rating: 241 MVA)
DFAX: 7.8%
Pre-project loading: 159.4%
3. Facility: Swatt – Lawrence Hill 100 KV
Contingency: Sweatt – Tinley Park 100 KV
Loading: 148.8% (Rating: 229 MVA)
DFAX: 6.1%
Pre-project loading: 146.0%
4. Facility: S Manheim – Dairyman's 100 KV Cir. 1
Contingency: South Manheim – Dairyman's Substation 100 KV #2
Loading: 102.9% (Rating: 168 MVA)
DFAX: 8.1%
Pre-project loading: 97.7%
5. Facility: Goodman Energy Ctr – Forman 100 KV Circuit 1
Contingency: Forman – Goodman Energy Ctr 100 KV Ckt 2
Loading: 100.0% (Rating: 143 MVA)
DFAX: 26.4%
Pre-project loading: 80.3%

6. Facility: Goodman Energy Ctr – Forman 100 KV Line
Contingency: Goodman Energy Center – Forman 100 KV Cir. 1
Loading: 100.0% (Rating: 143 MVA)
DFAX: 26.4%
Pre-project loading: 80.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Arch Smith – Sweatt 100 KV	Rebuild line with higher-capacity conductor	\$1,812,000	4% (\$72,000)
Sweatt Substation – Arch Smith 100 KV Line 2	Rebuild line with higher-capacity conductor	\$1,242,000	4% (\$50,000)
Lawrence Hill – Sweatt 100 KV L	Rebuild line with higher-capacity conductor	\$1,860,000	3% (\$60,000)
S Manheim – Dairymans 100 KV Cir. 1	Reconductor line	\$2,744,000	76% (\$2,076,000)
Goodman Energy Ctr – Forman 100 KV Circuit 1	Reconductor line	\$452,000	100% (\$452,000)
Goodman Energy Ctr – Forman Substation 100 KV Cir. 2	Reconductor line	\$499,000	100% (\$499,000)
	Total allocated to project		\$3,209,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.